



成都信息工程大学
Chengdu University of Information Technology

计算机图形学-第3章 开发环境和编程基础

Computer Graphics

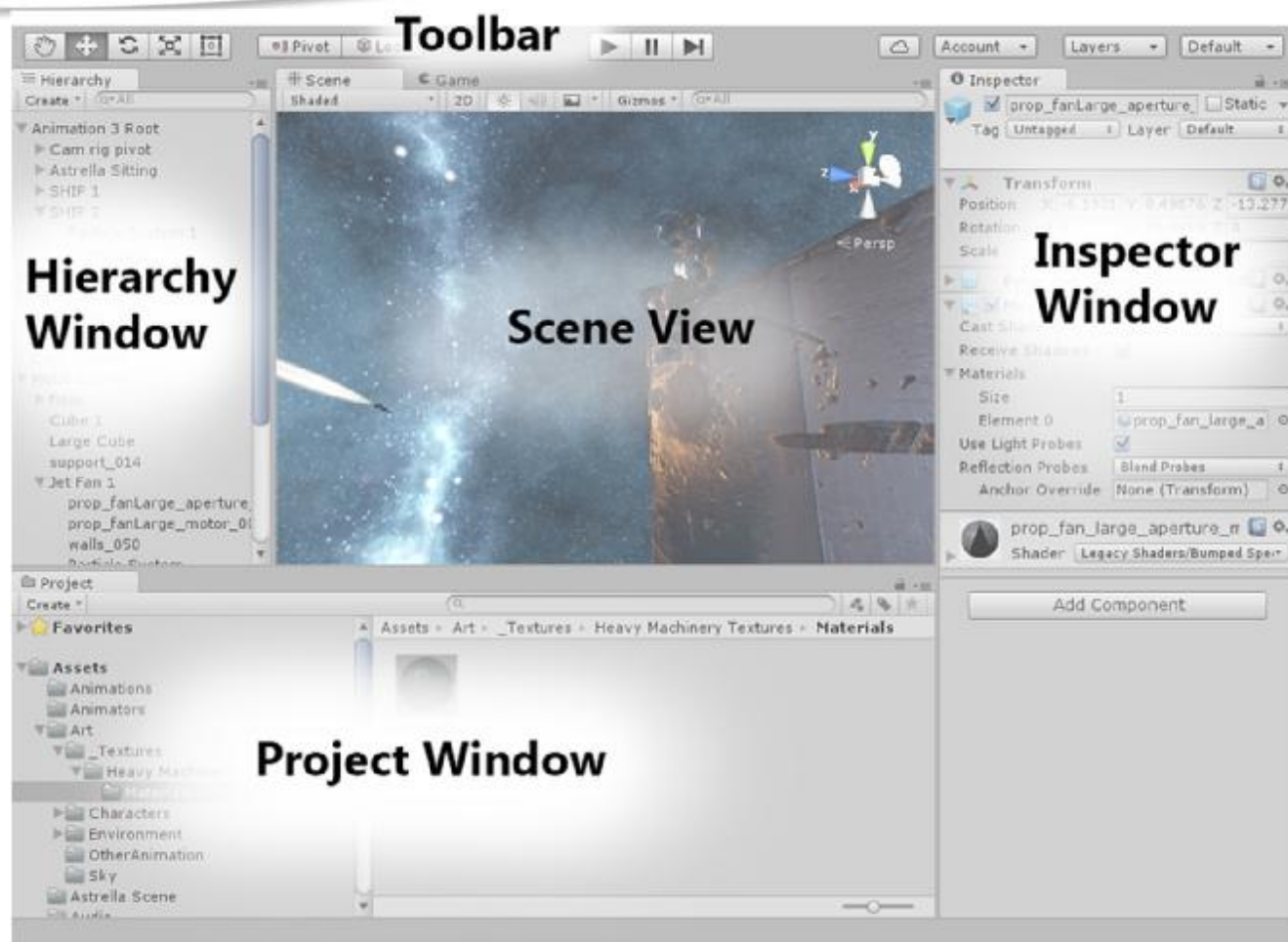
第三章 开发环境和编程基础



第三章 开发环境和编程基础

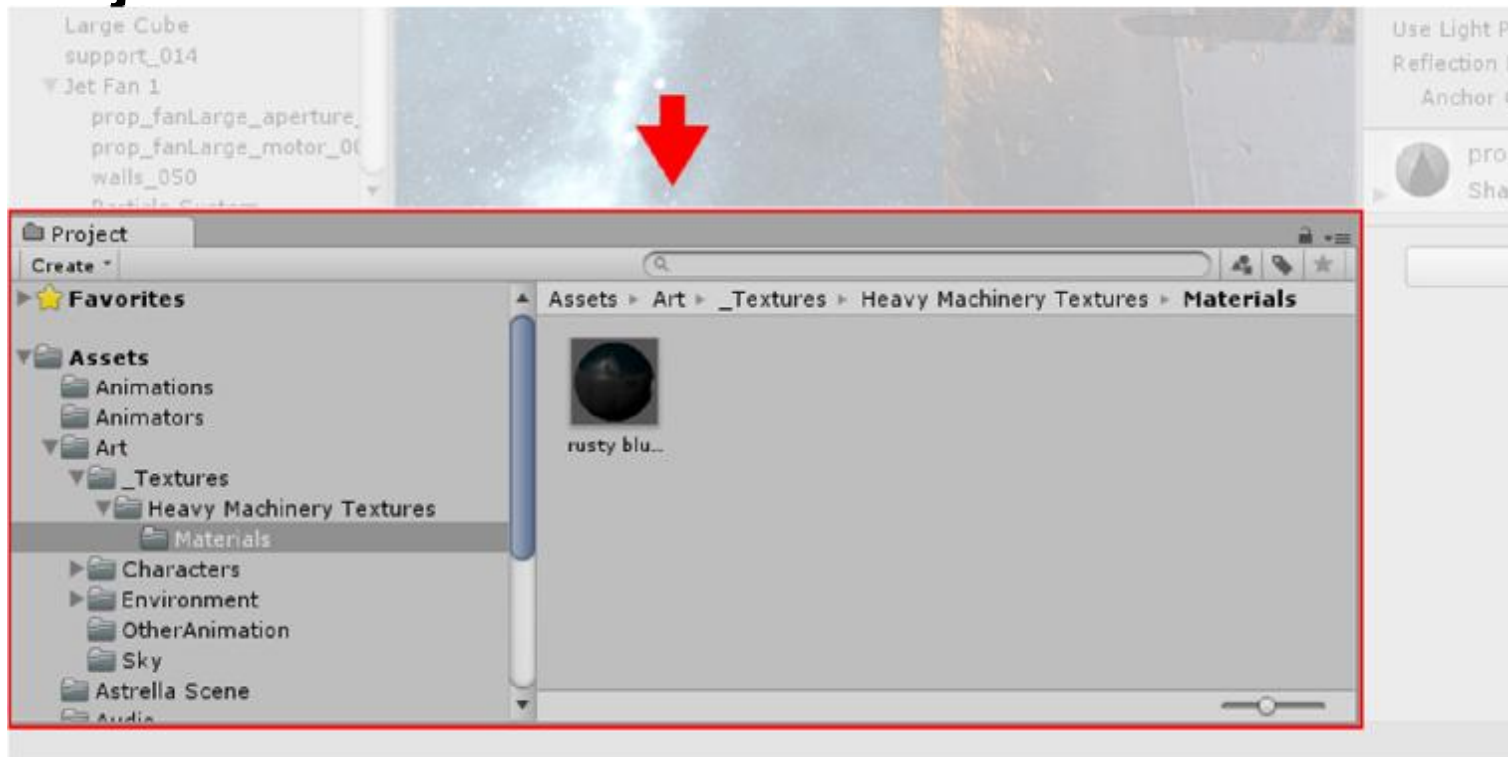
-  一、Unity3D 2019
-  二、The Main Windows
-  三、开发过程
-  四、Scripting
-  五、Graphics Hello World!

—、Unity3D 2019



一、Unity3D 2019

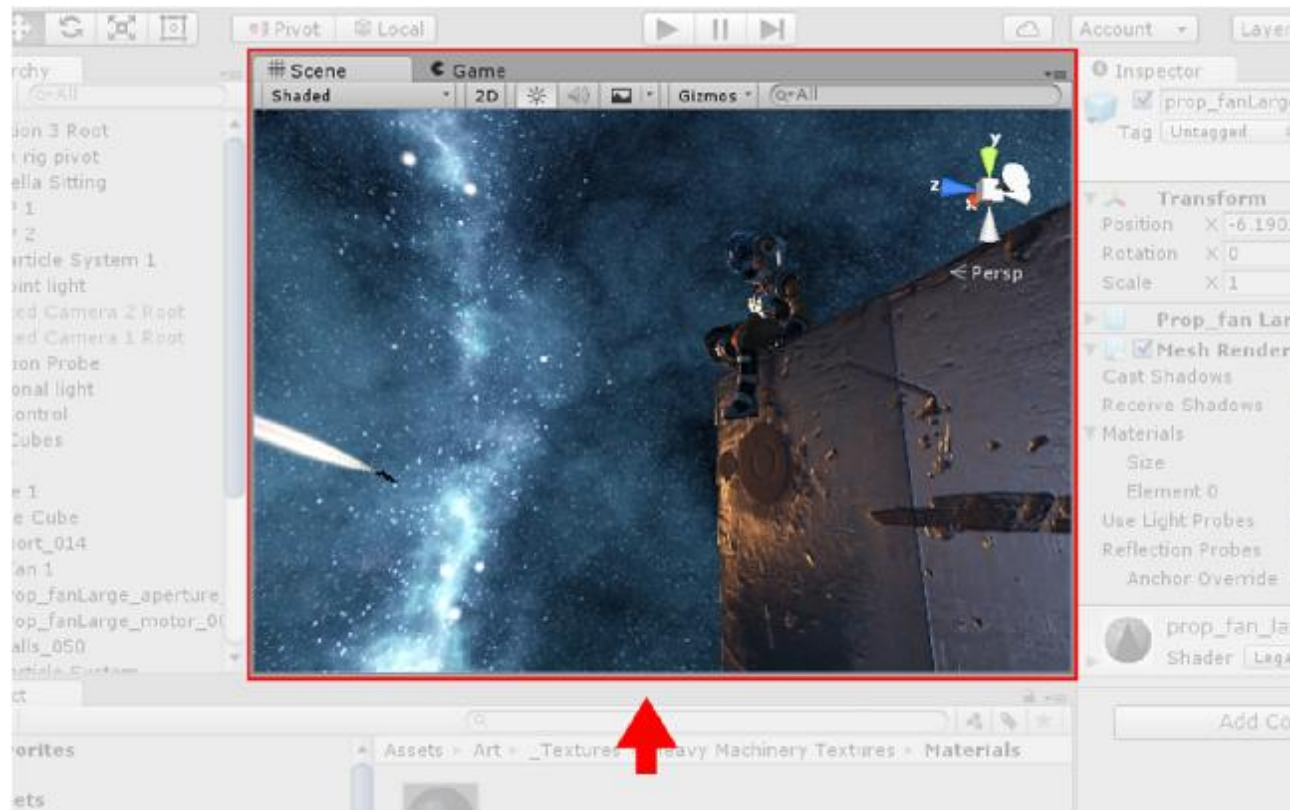
n The Project Window



The Project Window displays your library of assets that are available to use in your project. When you import assets into your project, they appear here.

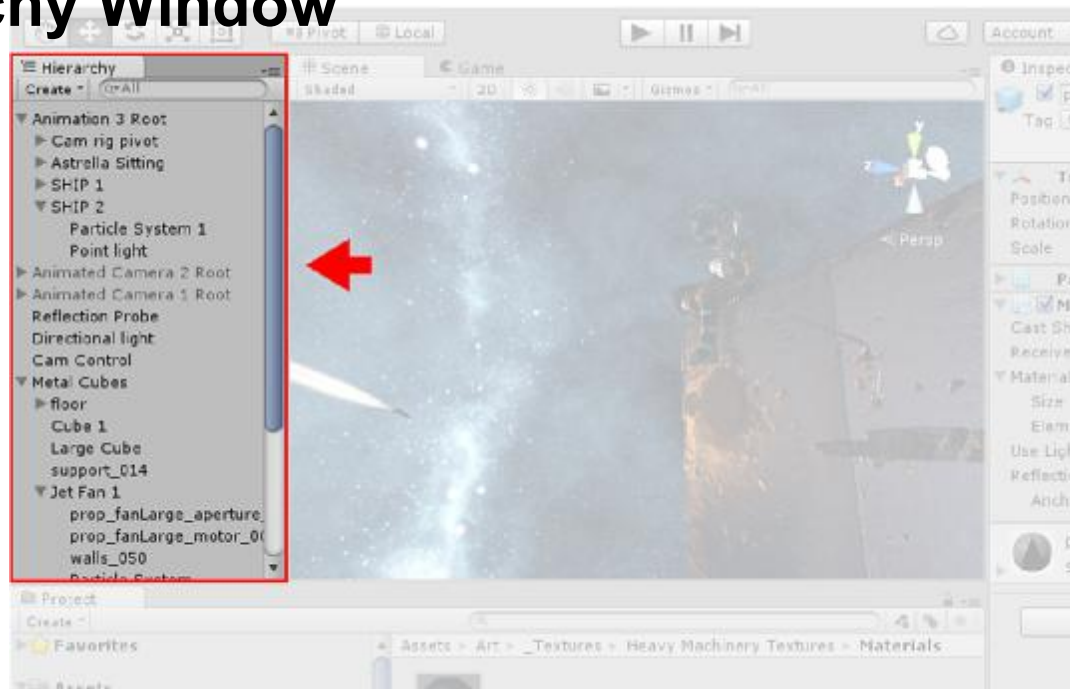
一、Unity3D 2019

n The Scene View



一、Unity3D 2019

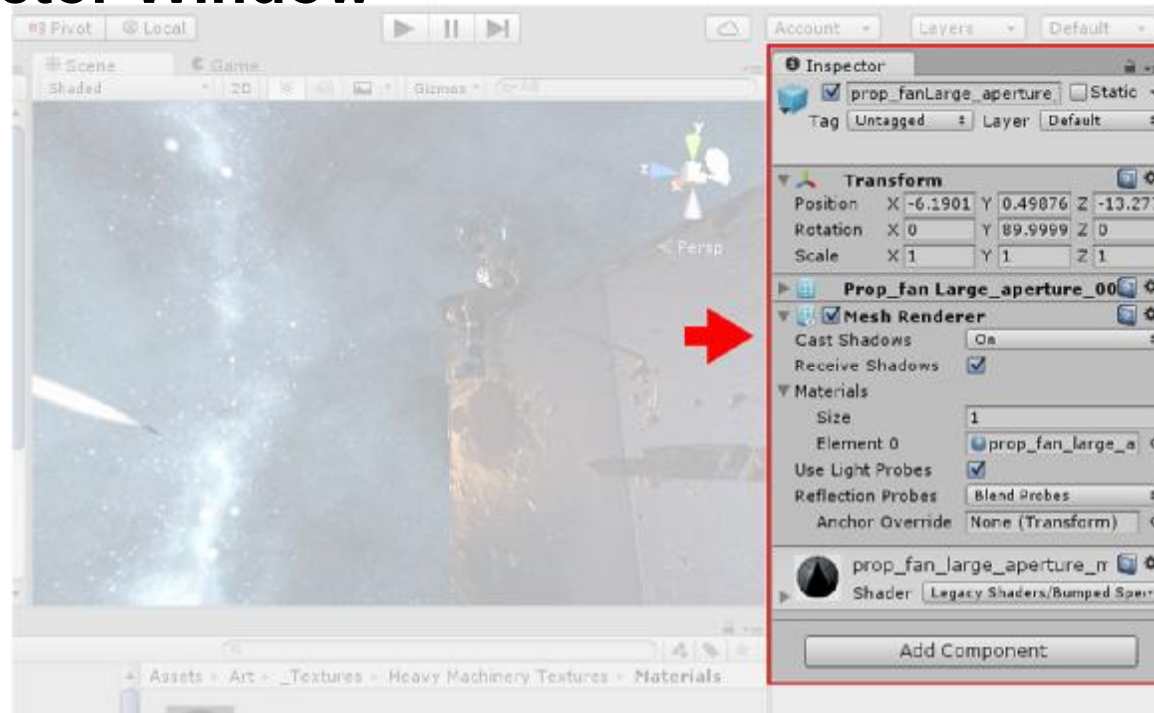
n The Hierarchy Window



The Hierarchy Window is a hierarchical text representation of every object in the scene. Each item in the scene has an entry in the hierarchy, so the two windows are inherently linked. The hierarchy reveals the structure of how objects are attached to one another.

一、Unity3D 2019

n The Inspector Window



The Inspector Window allows you to view and edit all the properties of the currently selected object. Because different types of objects have different sets of properties, the layout and contents of the inspector window will vary.

一、Unity3D 2019

n The Toolbar



The Toolbar provides access to the most essential working features. On the left it contains the basic tools for manipulating the scene view and the objects within it. In the centre are the play, pause and step controls. The buttons to the right give you access to your Unity Cloud Services and your Unity Account, followed by a layer visibility menu, and finally the editor layout menu

二、The Main Windows

The Project Window

The Scene View

The Hierarchy Window

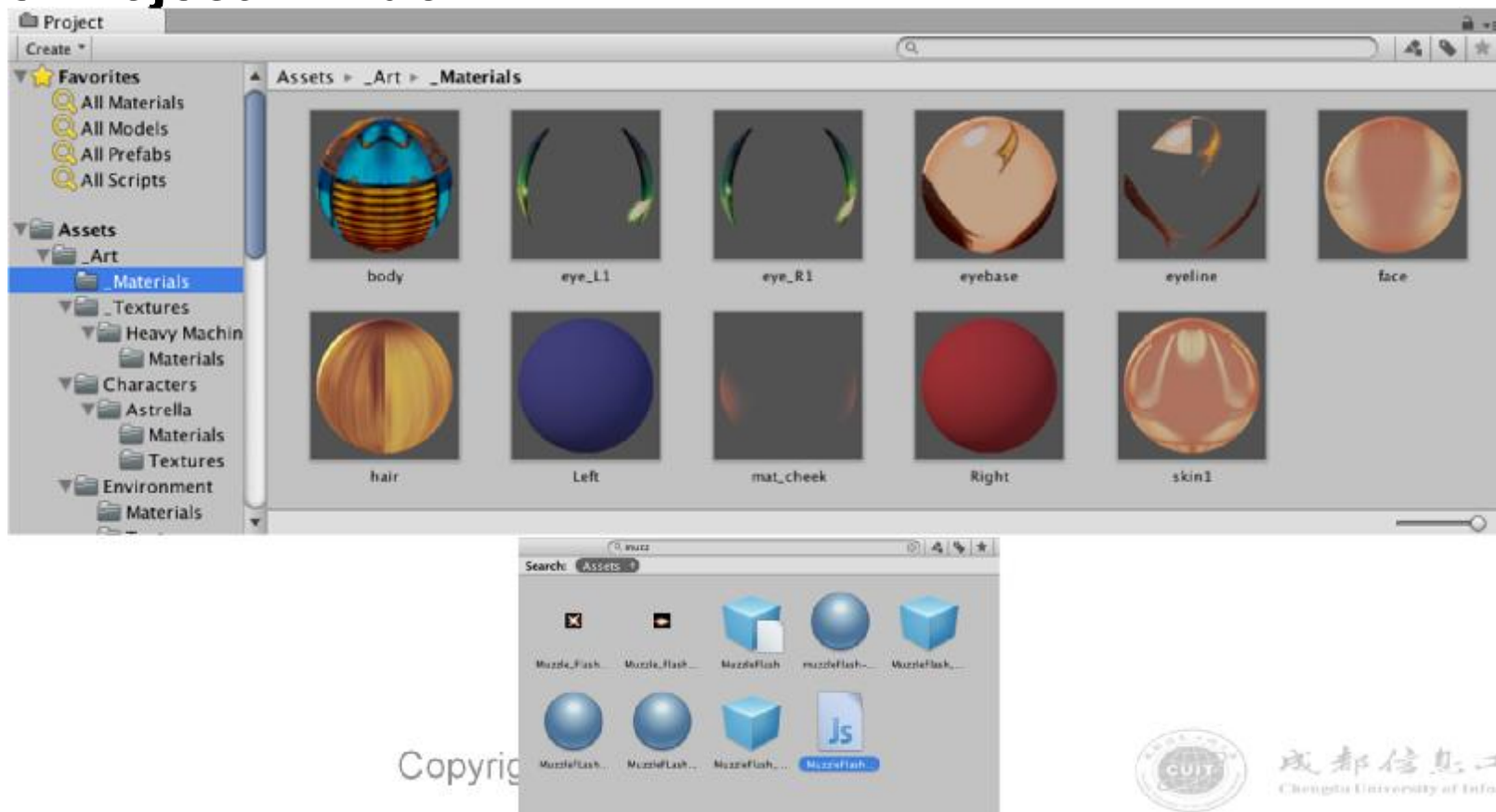
The Inspector Window

The Toolbar

The Game View

二、The Main Windows

n The Project Window



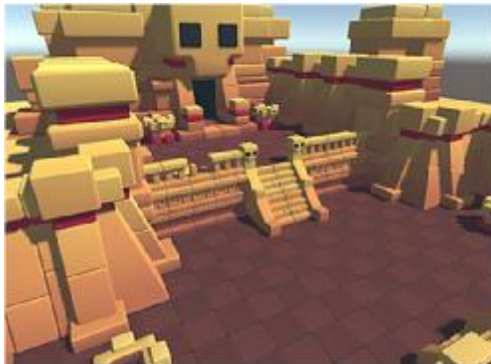
二、 The Main Windows

n The Scene View-Scene Gizmo



n The arms at the forefront are labelled **X**, **Y** and **Z**. Click on any of the conical axis arms to snap the Scene View Camera to the axis it represents.

n You can also toggle **Perspective** on and off. This changes the projection mode of the Scene View between **Perspective** and **Orthographic**.



二、 The Main Windows

n Moving, orbiting and zooming in the Scene View



n **Move:** Click-drag to drag the Camera around.



n **Orbit:** Hold Alt +click, and drag to orbit the Camera around the current pivot point. This option is not available in 2D mode, because the view is orthographic.

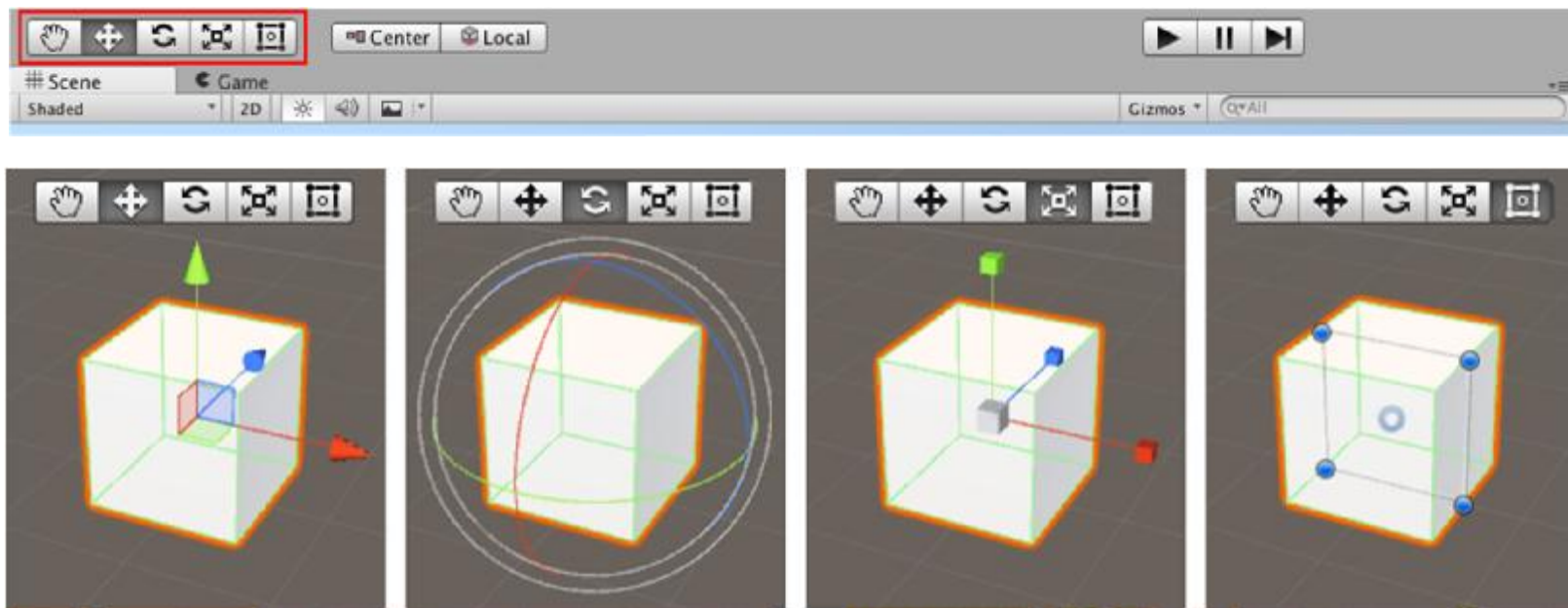


n **Zoom:** Hold Alt +right click, and drag to zoom the Scene View. On Mac you can also hold Control + click and drag instead.

二、The Main Windows

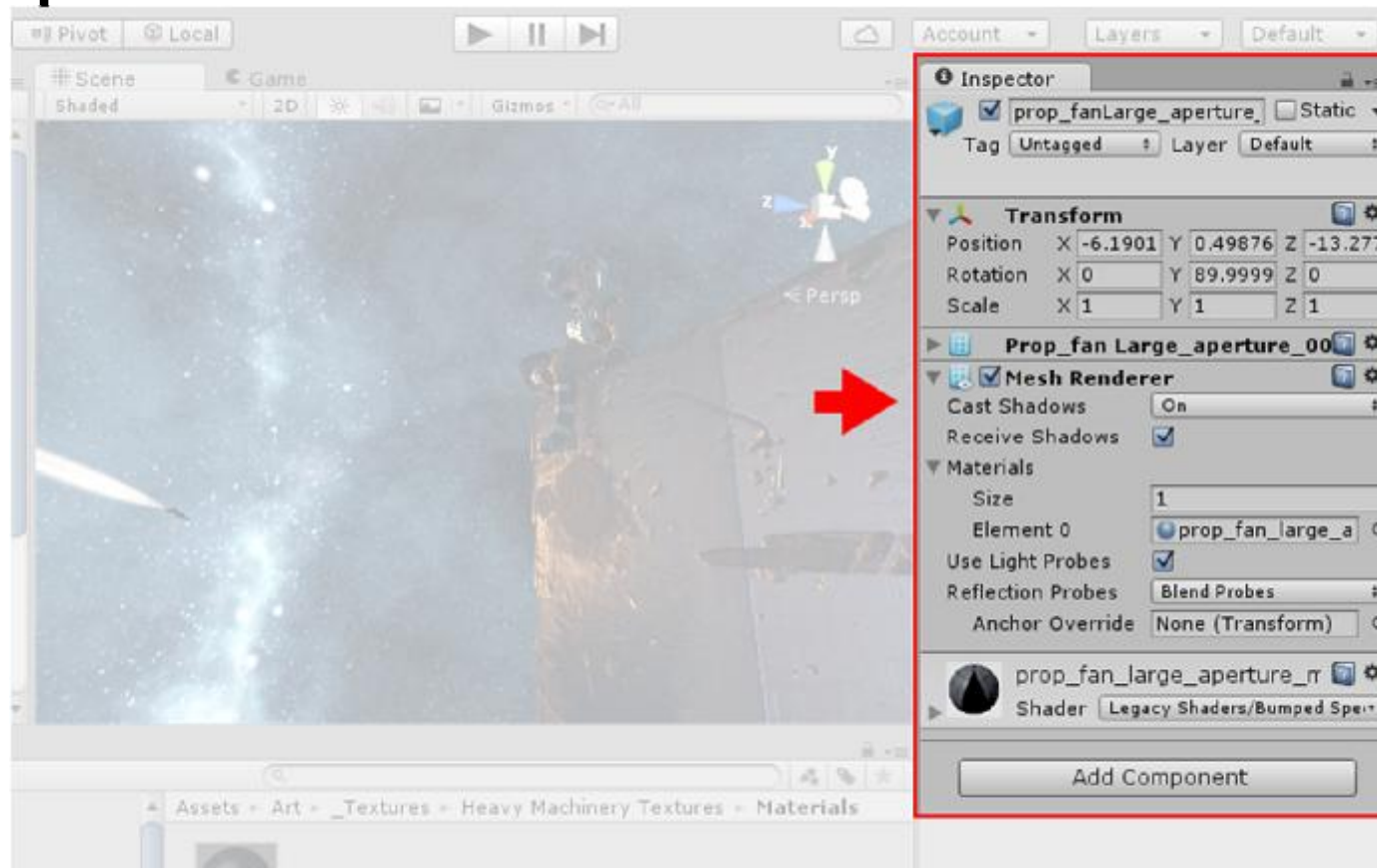
n Positioning GameObjects

n Move, Rotate, Scale, and RectTransform



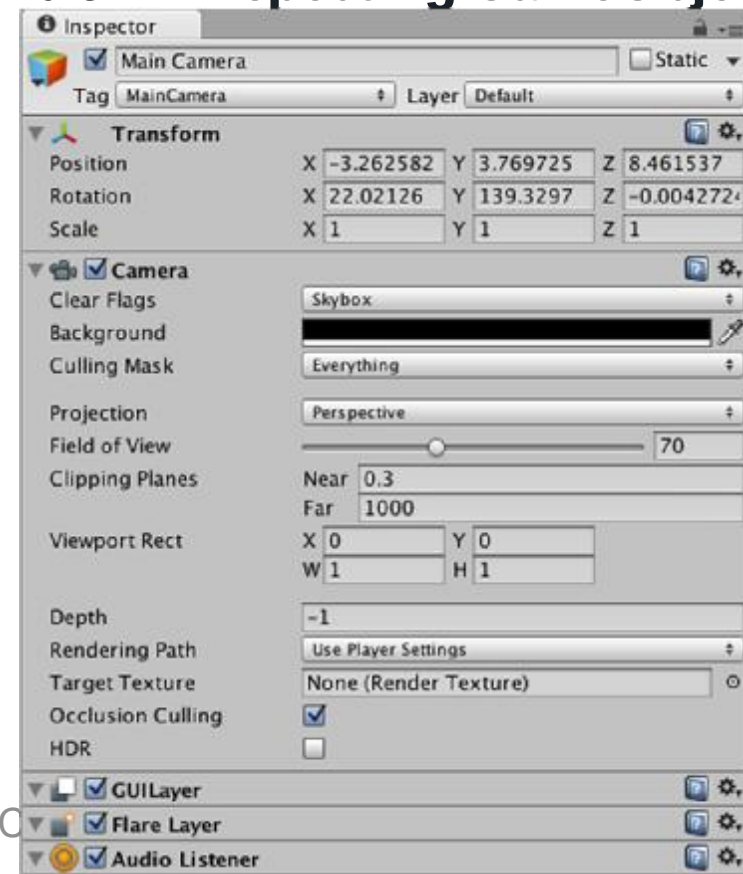
二、The Main Windows

n The Inspector window



二、The Main Windows

n The Inspector window- Inspecting GameObjects



二、 The Main Windows

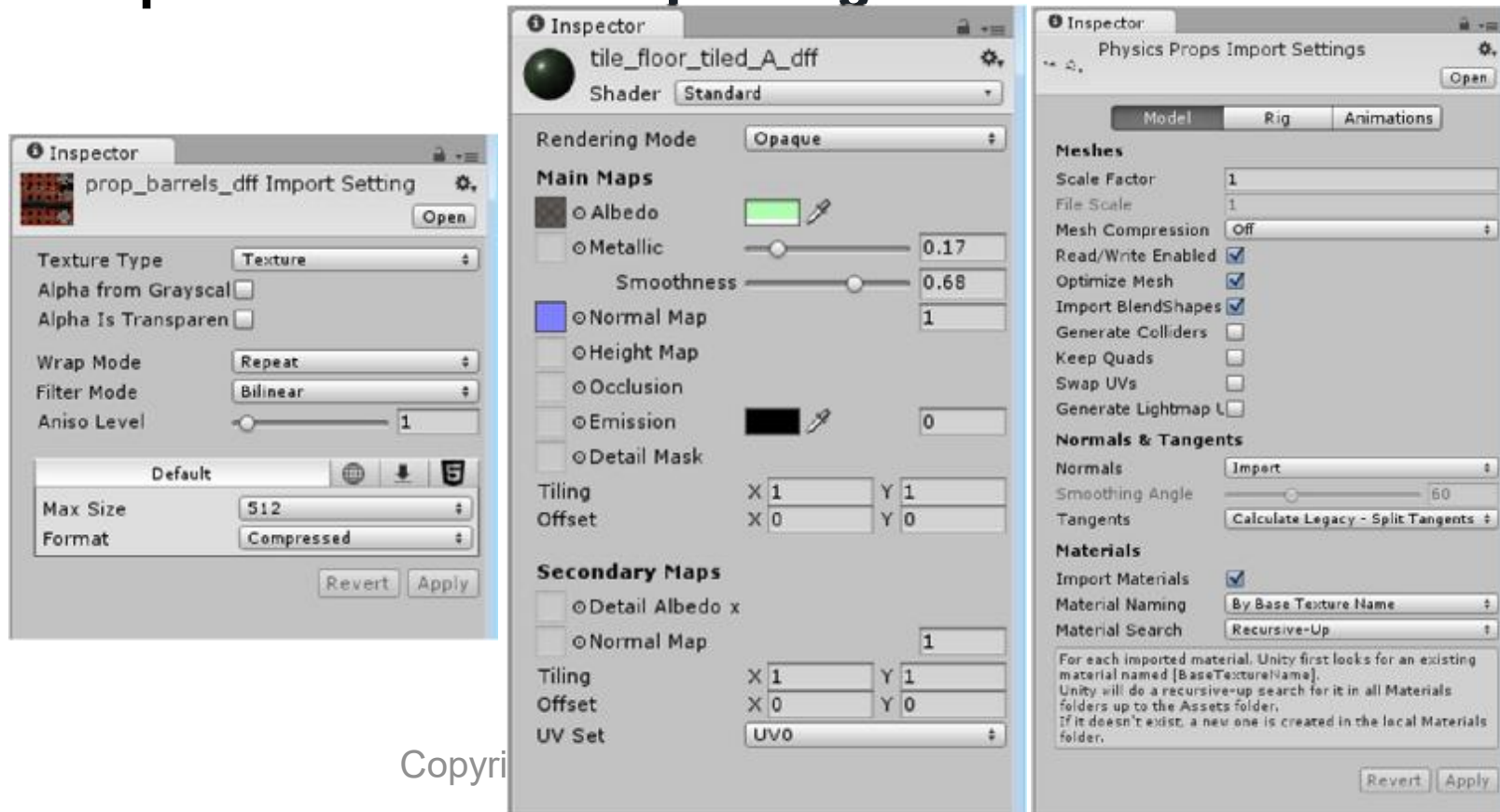
n The Inspector window

- Inspecting script variables



二、 The Main Windows

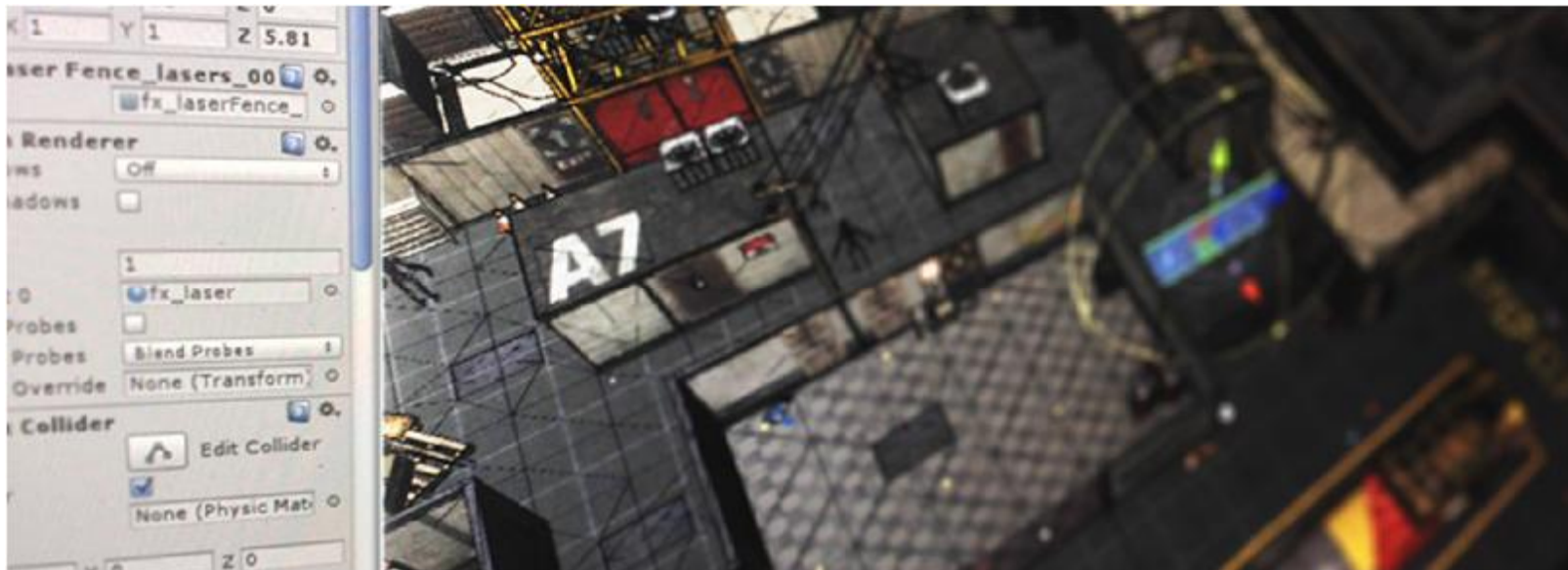
n The Inspector window- Inspecting Assets



三、开发过程

n Scenes

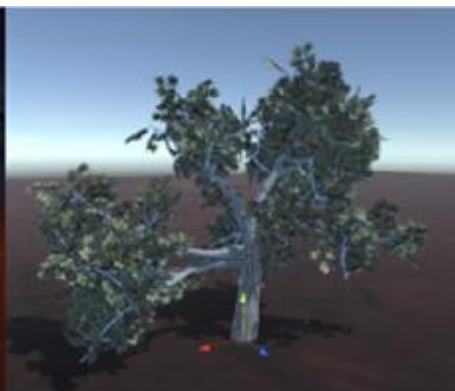
- n Scenes contain the environments and menus of your game.
- n A scene is a world that you created.



三、开发过程

n GameObjects

n Every object in your scene is a **GameObject**



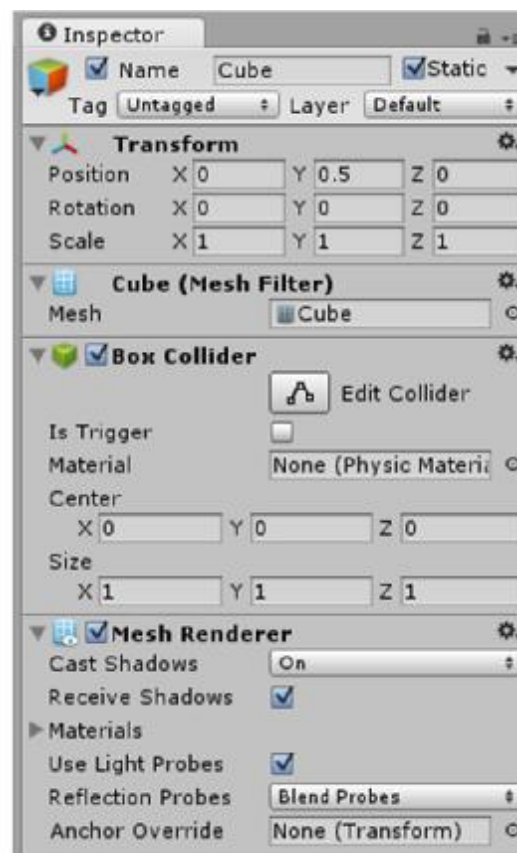
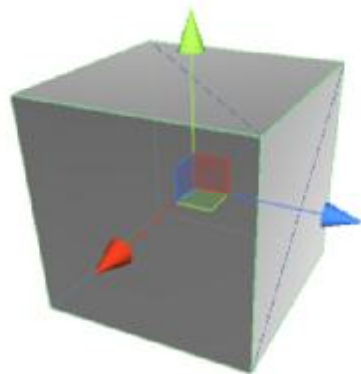
三、开发过程

n GameObjects



三、开发过程

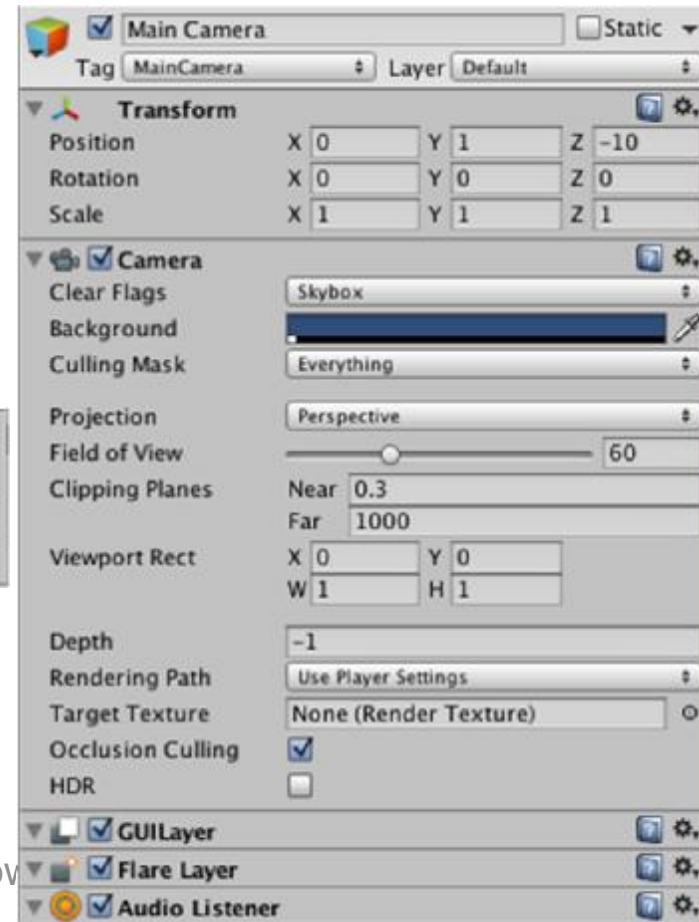
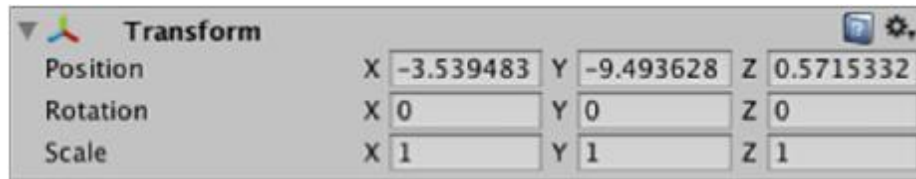
n GameObjects



三、开发过程

n Components (组件)

n A **GameObject** contains components.



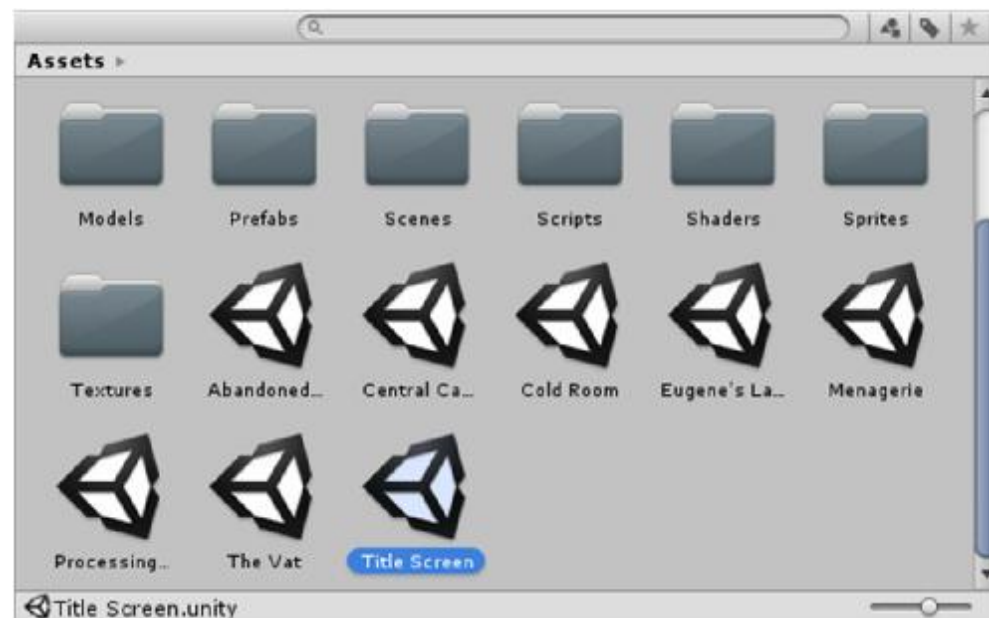
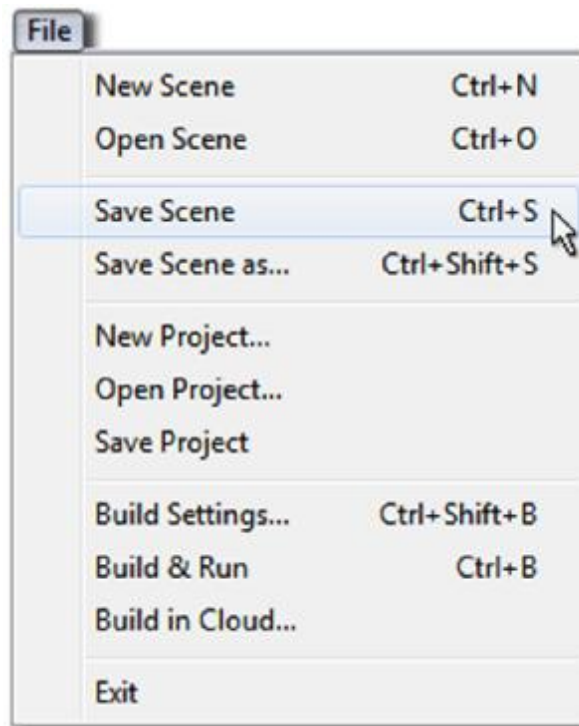
三、开发过程

n Prefabs (预制体)

- n Unity has a Prefab **asset type** that allows you to store a GameObject complete with components and properties.
- n The prefab acts as a **template** from which you can create new object **instances** in the scene.

三、开发过程

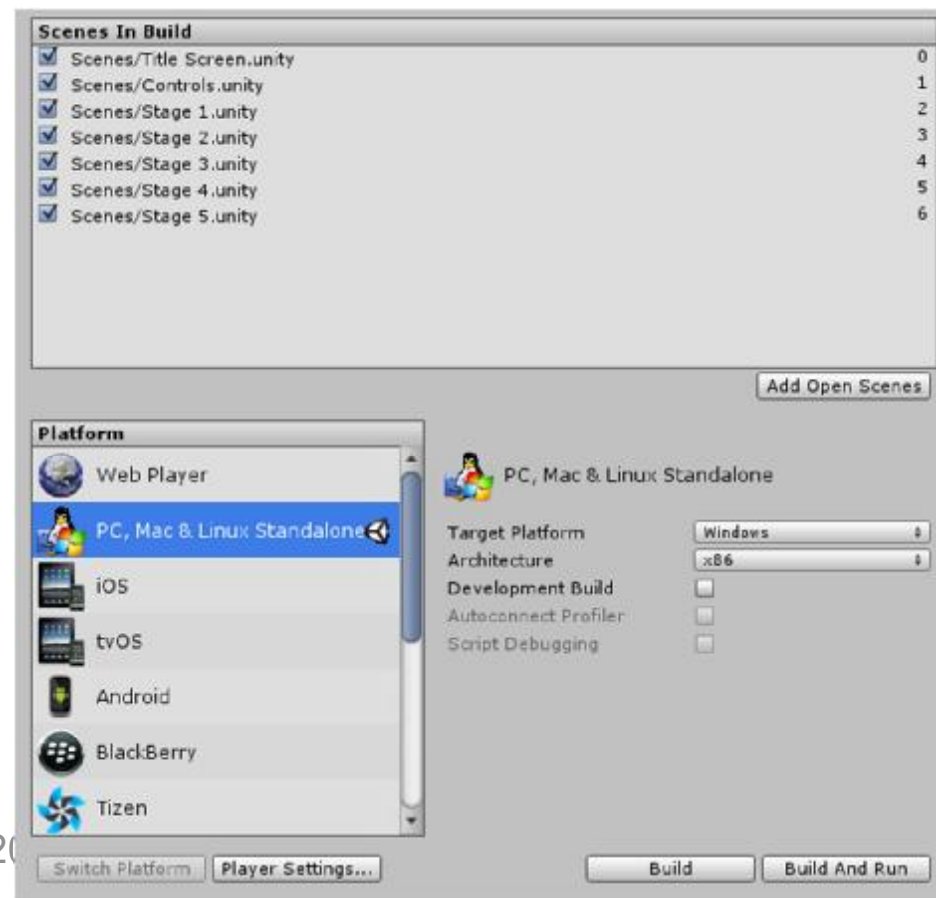
n Saving Your Work



三、开发过程

n Publishing Builds

n File->Build Settings...



四、Scripting

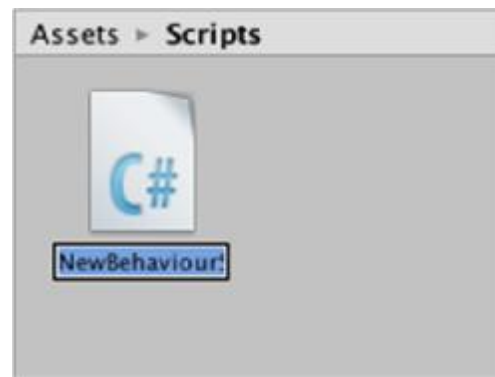
n We use C# scripting.



四、Scripting

n Creating Scripts

n Assets > Create > C# Script...



四、Scripting

n Anatomy of a Script file

```
using UnityEngine;
using System.Collections;
public class MainPlayer : MonoBehaviour
{
    // Use this for initialization
    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {

    }
}
```

四、Scripting

n Variables and the Inspector

```
using UnityEngine;
using System.Collections;
public class MainPlayer : MonoBehaviour
{
    public string myName;
    // Use this for initialization
    void Start()
    {
        Debug.Log("I am alive and my name is " + myName);
    }
    // Update is called once per frame
    void Update()
    {

    }
}
```



四、Scripting

- n MonoBehaviour

- n Execution Order of Event Functions

生命周期图

四、Scripting

n Event Functions

```
void Update()
{
    float distance = speed * Time.deltaTime *
Input.GetAxis("Horizontal");
    transform.Translate(Vector3.right * distance);
}
void FixedUpdate()
{
    Vector3 force = transform.forward * driveForce *
Input.GetAxis("Vertical");
    rigidbody.AddForce(force);
}
void LateUpdate()
{
    Camera.main.transform.LookAt(target.transform);
}
```



五、Graphics Hello World!

Now, we will use Unity3D and Visual Studio tools to construct your first 3D world, and write down your first line of 3D C# code.



成都信息工程大学
Chengdu University of Information Technology

计算机图形学-第3章 开发环境和编程基础

结束!